

1. What module can we use to run BLAST over the internet in Biopython:

1 point

- ☒ 1 `Bio.Blast.NCBIWWW`
- ☐ 1 `WWW`
- ☐ 1 `Bio.Blast.NCBIXML`
- ☐ 1 `NCBIXML`

2. Which one of the following modules is not part of the Bio.Blast package in Biopython:

1 point

- ☒ 1 FastaIO
- ☐ 1 ParseBlastTable
- ☐ 1 NCBIXML
- ☐ 1 NCBIStandalone

3. Using Biopython find out what species the following unknown DNA sequence comes from:

1 point

```
1 TGGGCCTCATTTTATCCTATATACCATGTTTCGTATGGTGGCGCGATGTTCTACGTGAATCCACGTTCTGAAGGACATCATACCAA
2 AATTAGGACCTCGATATGGTTTTATTCTGTTTATCGTATCGGAGGTTATGTTCTTTTGGCTCTTTTCGGGCTTCTCTCATTCT
3 CTACGGTAGAG
```

Hint. Identify the alignment with the lowest E value.

- ☒ Nicotiana tabacum
- ☐ Salvia miltiorrhiza
- ☐ Antirrhinum majus
- ☐ Citrullus lanatus

4. Seq is a sequence object that can be imported from Biopython using the following statement:

1 point

from Bio.Seq import Seq

If my_seq is a Seq object, what is the correct Biopython code to print the reverse complement of my_seq?

Hint. Use the built-in function help you find out the methods of the Seq object.

- ☒

```
1 print('reverse complement is %s' % my_seq.reverse_complement())
```
- ☐

```
1 print('reverse complement is %s' % complement(my_seq.reverse()))
```
- ☐

```
1 print('reverse complement is %s' % my_seq.reverse())
```
- ☐

```
1 print('reverse complement is %s' % reverse(my_seq.complement()))
```

5. Create a Biopython Seq object that represents the following sequence:

1 point

```
1 TGGGCCTCATATTTATCCTATATACCATGTTTCGTATGGTGGCGCGATGTTCTACGTGAATCCACGTTTCAAGGACATCATACCAA  
2 AATTAGGACCTCGATATGGTTTTATTCTGTTTATCGTATCGGAGGTTATGTTCTTTTTCGCTCTTTTCGGGCTTCTTCTATTCT  
3 CTACGGTAGAG
```

Its protein translation is:

☒ 1 WASYLSYIPCSYGGAMFYVNP~~RSK~~DIIPKSYN*DLD~~M~~VLFCLSYRRLCSFLLFFGLLLILLWHLR*

☐ 1 *FWPHIYPIYHVRMVARCSTSIHVRRTSYQ~~S~~RTIRTSIW~~F~~YCLSYRRLCSFLLFFGLLLILLWHLR

☐ 1 ILASYLSYIPCSYGGAMFYVNP~~RSK~~DIIPKSYN*DLD~~M~~VLLFIVSEVMFFALFRASSHSSLAPT~~V~~

☐ 1 FWPHIYPIYHVRMVARCSTSIHVRRTSYQ~~S~~RTIRTSIW~~F~~YCLSYRRLCSFLLFFGLLLILLWHLR*